18.Stock Buy and Sell – Max 2 Transactions Allowed

```
#include <bits/stdc++.h>
using namespace std;

class Solution {
  public:
    int maxProfit(vector<int> &prices) {
        int n = prices.size();
       if (n < 2) return 0
        vector<int> left(n, 0), right(n, 0)
        // left[i] = max profit for [0..i] with at most one transaction
        int minPrice = prices[0];
        for (int i = 1; i < n; ++i) {
            minPrice = min(minPrice, prices[i]);
            left[i] = max(left[i-1], prices[i] - minPrice);
        }   // right[i] = max profit for [i..n-1] with at most one transaction
        int maxPrice = prices[n-1];
        for (int i = n-2; i >= 0; --i) {
            maxPrice = max(maxPrice, prices[i]);
            right[i] = max(right[i+1], maxPrice - prices[i]);
        }

        // combine the two
        int ans = 0;
        for (int i = 0; i < n; ++i) {
            ans = max(ans, left[i] + right[i]);
        }
        return ans;
    }
};
```